

Low-Cost FPGA TDC With High Resolution and Density

Jiajun Zheng, Ping Cao, Di Jiang, and Qi An

Abstract—Time-to-digital converter (TDC) implemented in field programmable gate array (FPGA) has been developed for years and obtained superior performance. In an application with large amount of channels, tradeoff must be weighed between performance and cost if TDC is implemented in FPGA. Massive channels with high event rate also lead to big challenges for high data rate transmission in real time. Thus, it has great significance to develop time digitizers that accommodate high data rate with satisfying resolution and cost. In this paper, a TDC prototype with sandwich structure implemented in low-cost FPGA is presented. It contains 320 time measurement channels (ten TDC daughter cards) and can be used as digitizer for evaluating super module detectors in compressed baryonic matter time-of-flight experiment. Gigabit transceiver integrated in FPGA is utilized to support over 1-Gb/s data rate of a single TDC daughter. Preliminary tests show rms of leading edge measurement is better than 15 ps, and that of TOT measurement implemented by a single channel is better than 28 ps.

Index Terms—Field programmable gate arrays (FPGAs), rms resolution, time measurement, time-to-digital converters (TDCs).

I. INTRODUCTION

TIME measurement is indispensable for particle identification in high-energy physical experiments, especially in time-of-flight (TOF) system [1]–[4]. Field programmable gate array (FPGA) and application specific integrated circuit (ASIC) are two main kinds of time-to-digital converter (TDC) implementation techniques at present. Compared with ASIC TDC, FPGA TDC is able to be reconfigured limitlessly and has a shorter development cycle, which can be comparative advantage in specific demands. After years of development, FPGA TDC has become fully fledged [5]–[12]. The prevailing implementation principle of most high-resolution FPGA TDCs is combining coarse time measurement with fine time measurement. Generally, a counter provides coarse resolution for easily adjusting dynamic range, and a tapped delay line is used to perform time interpolation within a coarse clock period to acquire better resolution.

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FPGA underlying resources (e.g., carry logic and lookup table in a slice of Xilinx's FPGA) are widely utilized to perform the interpolation. Some effective approaches like wave union launcher [13] or multichain measurements averaging method [14] can greatly improve time measurement resolution, but at the cost of longer dead time or more resource consumption compared to plain TDC. Nowadays, the ongoing upgrade of high-energy physical experiments have tendencies of more channels and higher event rate, e.g., compressed baryonic matter (CBM) experiment at the Facility for Antiproton and Ion Research. The CBM experiment aims at the phase diagram of strongly interacting matter at highest net baryon densities and moderate temperatures. TOF system is one of the key parts for hadron identification, and there are over ten thousand of channels with up to 100–300 kHz single-channel event rate, which brings a tremendous stress on real time data transmission of front-end electronics. Single-channel time resolution better than 25 ps and time over threshold (TOT) measurement ability are needed for the TDC of TOF [15].

In this paper, a prototype of low-cost FPGA TDC module with high resolution and density is presented. This module has a sandwich structure containing ten identical TDC daughter cards. Each TDC daughter supports up to 32 digitizing channels with TOT measurement, and it is implemented in a Xilinx low-end Artix-7 FPGA. The sandwich structure is customized for evaluating CBM-TOF super module detector and can be used for signal digitizing and reading out. TDC implementation methodology and prototype structure are introduced in Sections II and III, respectively. In Section IV, preliminary test results of the TDC are presented. In Section V, there are some discussions. The conclusion is drawn in Section VI.

II. TDC METHODOLOGY

A. Single-Channel TDC Implementation

The principle of time measurement process in a single channel is illustrated by Figs. 1 and 2. There are two crucial parts, i.e., carry chain, and coarse time counter, and other auxiliary blocks [12]. The counter tracks the coarse time elapsed once the TDC is activated. In practice, 250-MHz TDC system clock and 16-b counter are applied, so the coarse bin T_{CLK} is 4 ns and the dynamic range is 262 144 ns. Once the hit signal (represented by the rising edge) occurs at a channel, its falling edge is stretched and aligned to the next rising edge of running clock. The stretched pulse propagates along the carry chain freely, whose status (fine time information) can be recorded by a D flip-flop array in thermometer code

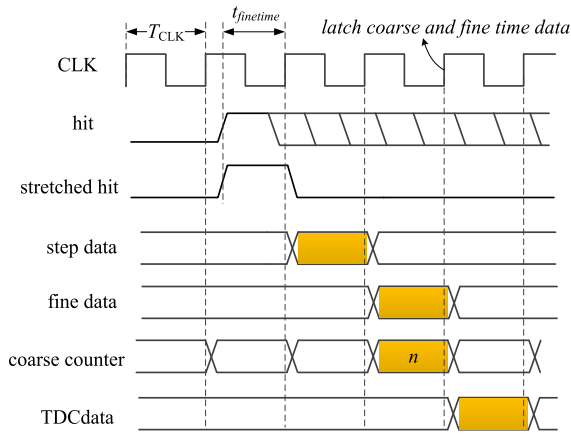


Fig. 1. Time diagram of time measurement.

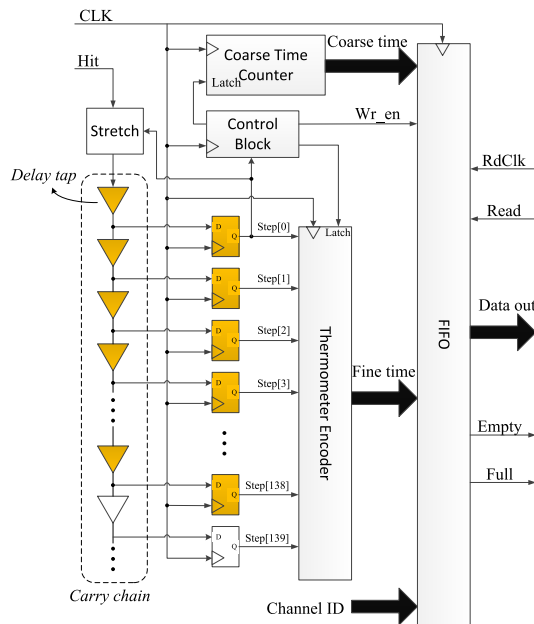


Fig. 2. Block diagram of single-channel TDC.

(step data in Fig. 1). Encoder based on dichotomy converts the thermometer code into binary code (fine data in Fig. 1) within one clock cycle. Control block latches both the coarse and fine time. Because of the logic delay for obtaining fine time data, the latching time is postponed to the third rising edge after hit occurs (see Fig. 1). As a result, the absolute time stamp t of a hit can be represented as following:

$$t = (n - 2) \cdot T_{CLK} - t_{finedata}. \quad (1)$$

In practice, the dedicated fast look ahead carry logic called CARRY4 is utilized as time interpolator. CARRY4 locates in the slice of Xilinx Artix-7 FPGA configurable logic block and can be cascaded to form a long carry chain. Fig. 3 shows how a CARRY4 logic is subdivided into two delay taps. Port CO0 and CO3 are two outputs for raw fine time data, which can be captured by adjacent D flip-flops (AQ and DQ). Port CIN and COUT are used for cascading to the next tap. CARRY4 logic is not subdivided medially so as to achieve

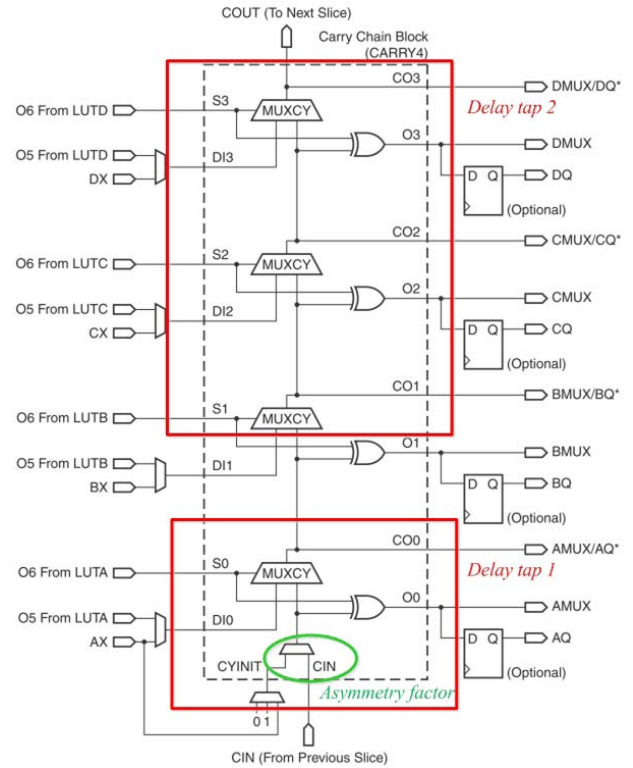


Fig. 3. Delay tap of carry chain implemented in CARRY4 logic. CO0 and CO3 are raw fine time data output ports. CIN and COUT are used for cascade.

better bin width uniformity because there is an asymmetry factor near CIN [16]. In a single TDC channel, 70 CARRY4 logics (140 delay taps) are cascaded vertically in FPGA fabric to form a carry chain, and the total delay is slightly more than one system clock cycle.

B. 32-Channel TDC Implementation

A 32-channel TDC capable of TOT measurement is implemented in Xilinx Artix-7 series FPGA. In tradition, to measure a pulsewidth, both of the leading and trailing edges of hit signal are detected in two channels, respectively. Thus the resource cost is doubled for one pulsewidth measurement. In fact, a single carry chain can also measure both two transition edges [16]. In Fig. 3, each XOR gate has one output port (O0–O3). If the input ports S0–S3 of CARRY4 are fed by high logic value “1,” reversed status value can be obtained along the carry chain. There are four D flip-flops in a slice, two of which have been used to record fine step data of leading edge, so there is only one delay tap that can be implemented in a CARRY4. As a result, similar to the case of leading edge, trailing edge can be measured if fine step data are output from O1 or O2 port along the carry chain.

The block diagram of TDC is shown in Fig. 4. It should be noted that if pulsewidth is measured by a single carry chain, the stretch block in Fig. 2 must be deleted in order to preserve the correct trailing edge. Therefore, pulsewidth must be wider than a clock cycle. Leading edge and trailing edge thermometer codes share the same encoder as they never occur at the same clock rising edge. If the leading edge and trailing

the synchronous clock and power of all TDC boards are also distributed by TRM through the golden fingers. Similar to TFB, TRM is also considered as mechanical structure for TDC daughters.

D. SFP Daughterboard

For the convenience of assembly, optical transceivers [small form-factor pluggable (SFP) or SFP+] used by TRM are placed on a separate daughterboard, which is also vertically plugged in between TFB and TRM by golden fingers similar to that of TDC daughters. Golden fingers support for data transmission and power supply as well. System synchronous clock can also be transmitted by optic fiber from back-end electronics to TRM through SFP daughterboard.

E. High-Speed Readout Link

In Fig. 5, digitized time data of all TDCs are gathered by TRM, and then transmitted through SFP daughterboard to back-end readout module. Optical transceivers are used to increased transmission speed and distance. For sufficient amount of gigabit transceivers and adequate line rate at an affordable cost, two Xilinx Kintex-7 FPGAs are utilized on TRM, one running in slave mode and the other in master mode. In each FPGA on TRM, five gigabit transceivers (called GTX) receive five TDCs' data, respectively, with 1.25-Gb/s line rate as calculated before. Among of these TDC daughters, five of them transmit data to slave FPGA and the others to the master one. The master aggregates all data and transmits to the back-end readout crates through SFP+ transceiver.

Fig. 6 shows the component photographs of the sandwich structure. The center area of TRM is hollowed out for inserting or extracting optical transceiver perpendicular to TRM. The customized mechanical metal cage provides all-around mechanical stabilized fixation for printed circuit boards (PCBs). Volume of the sandwich structure is approximately 12 cm × 15 cm × 42 cm.

IV. PRELIMINARY TEST AND VERIFICATION

A. Nonlinearity Characterization of TDC Channel

Statistical code density test [12] is used for performing characterization of nonlinearity. External test hits have a frequency of 338.7391 Hz and have no correlation with TDC system clock frequency. If there are N hits in total (N should be large) and these hits are assumed to evenly spread over a TDC system clock cycle 4000 ps, the width of an n -count bin is $n/N \times 4000$ ps. Wu [19] recommends bin-by-bin calibration since the widths of bins vary hugely, and the rms measurement error can be reduced. The calibrated bin edge t_n is written as following, where w_n is the n th bin width acquired by code density test

$$t_n = \frac{w_n}{2} + \sum_{k=0}^{n-1} w_k. \quad (3)$$

Fig. 7 shows the typical bin width test result of a single channel for the TDC daughter. The uniformity is improved better after bin-by-bin calibration. 120 delay taps are used in

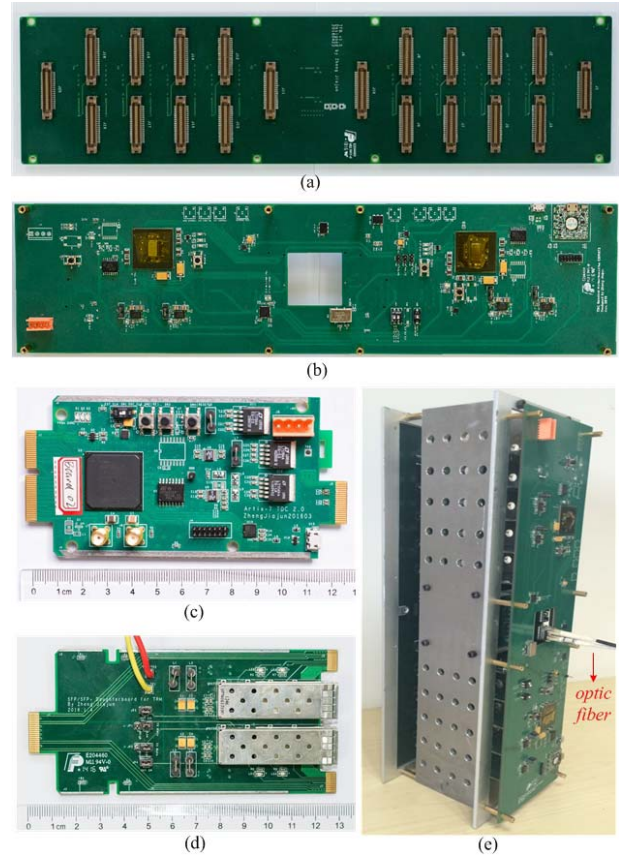


Fig. 6. Photographs of sandwich structure. (a) TFB. (b) TRM. (c) 32-channel FPGA TDC board. (d) SFP daughterboard. (e) Outside view of sandwich structure with optic fiber plugged.

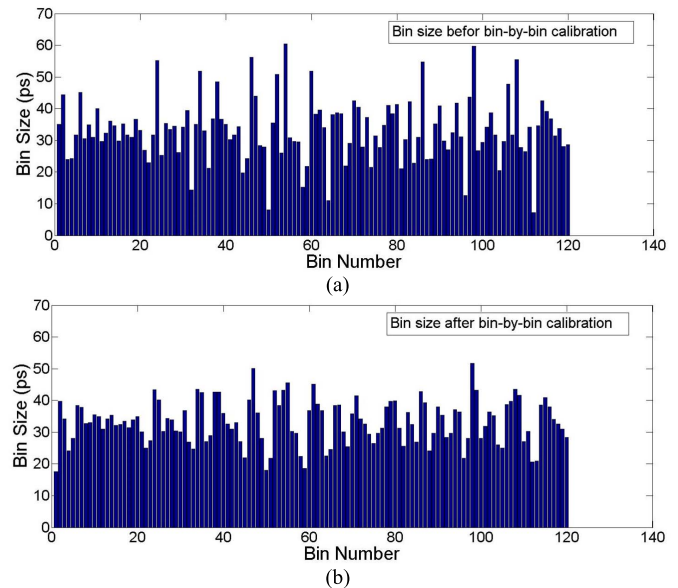


Fig. 7. Statistical analysis widths of TDC bins (a) before and (b) after bin-by-bin calibration.

a TDC channel according to the statistical result. As TDC system clock cycle is 4 ns, an approximate 33-ps average bin size can be obtained. A typical result of nonlinearity characterization of TDC channel is shown in Fig. 8 after bin-by-bin calibration. The differential nonlinearity is within 0.6 LSB, and

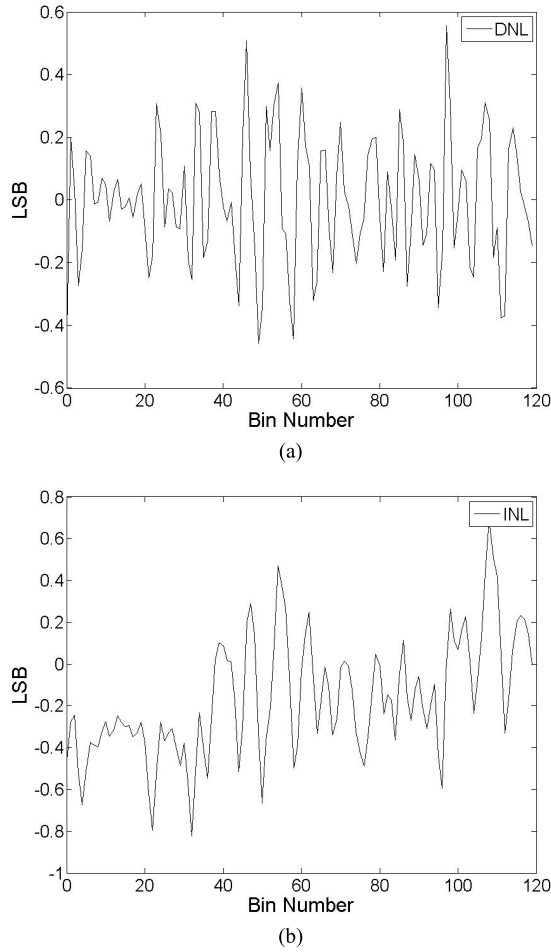


Fig. 8. (a) Differential and (b) integral nonlinearity of a single TDC channel after bin-by-bin calibration.

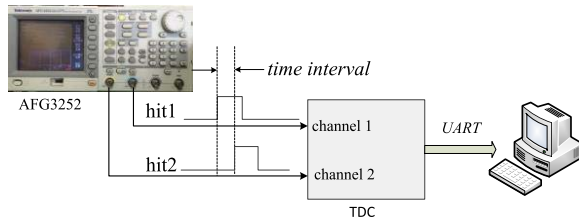


Fig. 9. Schematic of TDC rms test method.

the integral nonlinearity is within 0.8 LSB. The nonlinearity is mainly caused by the nonuniformity of the taps in carry chain and uneven clock distribution in the register structure. Process of FPGA fabric plays a crucial role in uniformity of dedicated carry chain structure. Even different FPGAs of same model can also vary with each other.

B. Time Measurement Performance

1) *Leading Edge Measurement*: As depicted in Fig. 9, two channels of hit signals generated by generator (Tektronix AFG3252) are fed into two TDC channels, respectively. Time interval of the two hits is measured by two TDC channels and the measurement data are sent to PC for analysis. If the

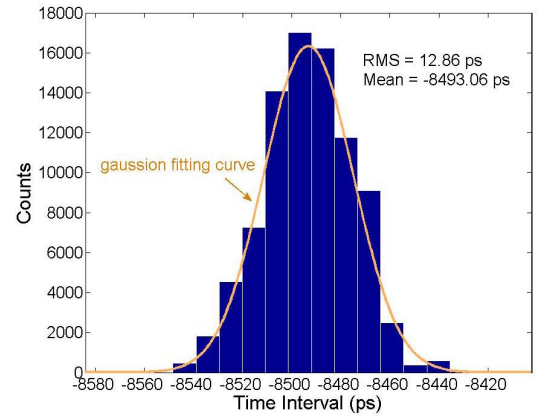


Fig. 10. Statistical spread of 8.5-ns time interval.

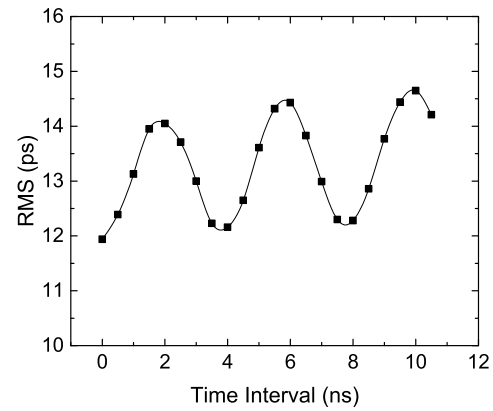


Fig. 11. RMS test result of one single TDC channel. RMS changes on a 4-ns cycle.

two channels are assumed to be identical, a single-channel rms resolution equals to standard deviation of the time interval divided by $\sqrt{2}$. Fig. 10 shows a spread of 8.5-ns time interval and the rms is 12.86 ps. Fig. 11 plots the statistic rms resolution of one single TDC channel with different time intervals, and the time intervals vary by a 0.5-ns step.

As two channels need to be utilized in order to obtain a single-channel rms resolution, every two channels of the 32-channel TDC are tested and a comprehensive rms statistics of each channel around any other one is shown in Fig. 12. Time interval between all channels is set to 0 ns. A 15-ps reference plane is drawn in Fig. 12. It turns out that all channels have an rms performance better than 15 ps.

Furthermore, rms resolution across different TDC daughter cards is also tested. As shown in Fig. 13, TRM provides synchronous 100-MHz clock and common start signal to two separate TDC cards through golden fingers in sandwich structure. Two channels of hits generated by AFG3252 are fed into two TDC daughters, respectively. Time measurement results of two TDC are transmitted to TRM and then sent to PC for analysis. Fig. 14 plots spread of 0-ns time interval between the two TDCs. Obviously, about 14-ns rms resolution is achieved successfully. This result is a little worse than that of one card in Fig. 10, mainly because of the skew and different jitter behaviors between synchronous clocks.

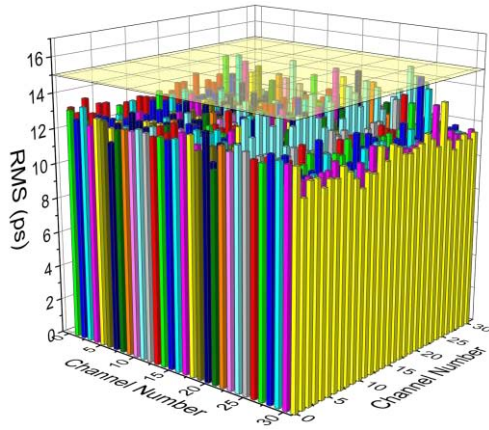


Fig. 12. RMS statistics of 32-channel TDC leading edge measurement. All values are below the plane of 15 ps.

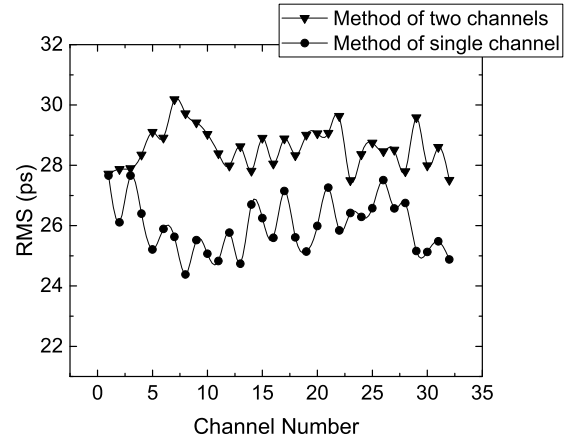


Fig. 15. RMS statistics of 32-channel TDC TOT measurement.

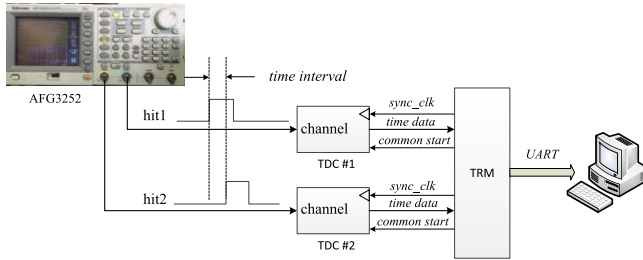


Fig. 13. Schematic of rms test across different TDC cards.

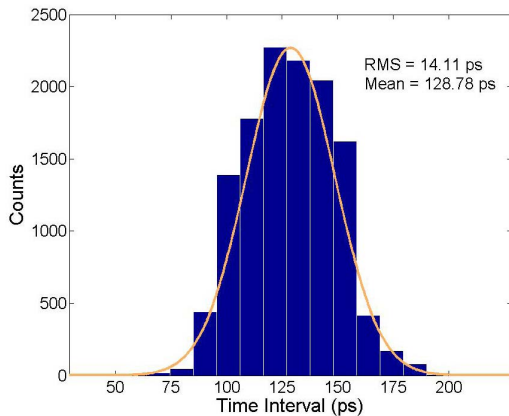


Fig. 14. Statistical spread of 0-ns time interval across two separate TDC cards.

2) *TOT Measurement*: As mentioned before, there are two methods to implement TOT measurement. The rms statistics of 32-channel TDC are compared in Fig. 15. Apparently, rms of TOT measurement implemented by a single channel is better than that of double channels. In fact, if a separate channel is used to measure trailing edge, it is too crowd to constrain all the locations of 64 carry chains in FPGA. The additional 32 carry chains are automatically routed by the software and the uniformity of delay taps is badly affected because of the crowd and irregular routing net. Thus rms resolution of trailing edge measurement will be worse than that of leading edge. TOT measurement with one single carry chain can obtain better resolution and less resource consumption.

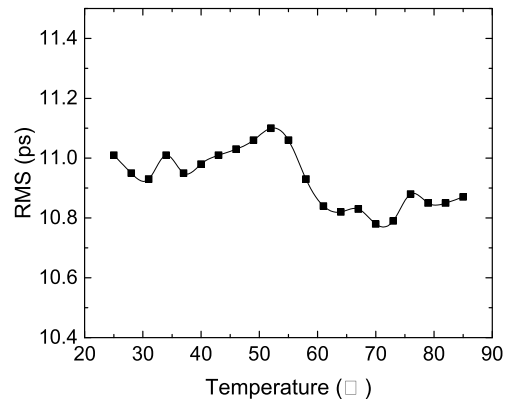


Fig. 16. Dependence of rms on temperature of FPGA.

C. Temperature Variation

TDC daughter card is placed in an electric thermostatic drying oven for temperature test. Time interval of hits fed into two channels is set to a fixed value of 0 ns. Temperature inside the oven is precisely controlled from 25 °C to 85 °C in 3 °C step. RMS test is performed at least ten minutes after the oven temperature is stable. Fig. 16 shows the rms variation along with temperature. It changes little although temperature varies a lot. The actual bin number of a carry chain changes from 120 to 123. It is inferred that temperature variation has limited impact on time resolution of this proposed FPGA TDC.

V. DISCUSSION

A. Bubble Suppression of Thermometer Encoder

Bubble suppression is crucial for encoder design. In an ideal case of a single TDC channel, the thermometer code recorded by registers array is like 111111000, where the low bits on the left are all logic “1.” There is one 1-0 transition normally and two or more bubbles rarely occur. A bubble can occur at the transition edge due to uneven propagation of delay taps, which causes the actual code may be 111101000.

The thermometer encoder is based on dichotomizing search principle. If the center bit of the thermometer code is “0,” the “1-0” transition is supposed to locate in first half of the

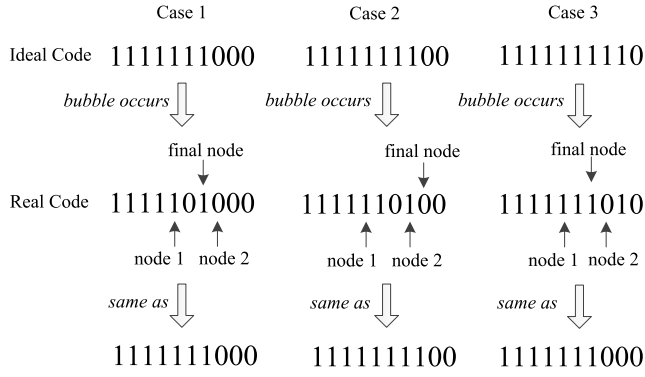


Fig. 17. Examples of encoding thermometer code with bubble based on binary search algorithm.

TABLE I
RESOURCE UTILIZATION AND OVERALL PERFORMANCE
OF 32-CHANNEL TDC

Resource consumption	Resource	Used	Available in FPGA
	Registers	12410	126800
	LUT	21378	63400
System clock frequency	250 MHz		
Number of delay taps	120		
Nonlinearity Characterization	DNL		± 0.6 LSB
	INL		± 0.8 LSB
RMS	Leading measurement		< 15 ps
	TOT measurement		< 28 ps
Dead time	10 ns		

code and vice versa. Taking the three cases in Fig. 17 for example, each real code has 10 b with one bubble compared to ideal code. The encoder needs to find three node bits to finish encoding, and the final node is designed to be searched from high-order bit near the last node. If the bubble does not occur on node bit, it will be neglected and the result is correct just like encoding ideal code (case 1 and case 2 in Fig. 17). However, if the bubble occurs on a node bit (case 3 in Fig. 17), the result will differ from the ideal case by value 2. In a way, this wrong encoding result contributes to time measurement error of TDC.

B. TDC Overall Performance

Table I shows the resource utilization and overall performance of the 32-channel TDC. Synchronous clock provided to TDC from TRM has a time interval error jitter of about 5-ps rms tested by oscilloscope independently, so the actual time measurement performance is considered to be better. Specification of 25-ps rms resolution on leading edge measurement of CBM-TOF TDC is totally met, and the TOT measurement resolution with 28 ps is acceptable.

In this proposed TDC prototype, dead time of TDC channel is about 10 ns. It is mainly affected by system clock cycle and the fine time measurement principle. Actually, taking the maximum event rate 300 kHz of CBM-TOF into consideration, the average time between two events is 3333 ns, so a 10-ns dead time is good enough to meet the requirement of TOF system.

C. TDC Cost

The components of 32-channel TDC mainly consist of a Xilinx low-end FPGA and several linear voltage regulators as well as some auxiliary small devices like buttons. The eight-layer PCB size is only about 13 cm $\times$ 6 cm [see Fig. 6 (c)]. So cost is kept to a minimum. According to the resource utilization in Table I, it is possible to choose XC7A75T-1FGG484 FPGA (less logic cells than XC7A100T-1FGG484 we used) to further reduce cost.

VI. CONCLUSION

Pivotal needs of time measurement in the TOF system of the ongoing upgraded high-energy physical experiment bring the necessity for TDC with high-density, high resolution, and low cost. In this paper, a 32-channel plain TDC implemented in a Xilinx low-end Artix-7 series FPGA is introduced. Based on the TDC basic module, a high-density prototype with sandwich structure is presented at the background of CBM experiment. This structure greatly reduces the number of connectors and cables, simplifies the distribution of synchronous clock and power, and significantly improves the system integration. The performance of the 32-channel TDC is evaluated. Result shows that this proposed method can meet the requirements of application.

It should be noted that although this high resolution and density TDC prototype is designed for CBM experiment TOF MRPC detector evaluation, it is easy to be modified and suits other high-energy physical experiments consisting of numerous channels.

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